Parsa Sinichi

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Summary

I am an M.Sc. student in Computer Engineering (AI and Robotics) at Ferdowsi University of Mashhad, working on interpretable AI. My research focuses on computer vision and medical image analysis, with an emphasis on retinal disease detection. I am also an AI R&D Engineer at C1Tech, where I develop voice assistants, speech recognition systems, and large language models, particularly for intent detection and slot filling in practical conversational interfaces. My background also includes robotics, particularly in developing software components and data-processing tools for lower-limb exoskeleton systems aimed at improving human–robot interaction.

Research Interests

- Medical Image Analysis
- Computer Vision
- Speech Recognition and Synthesis

Education

- **M.Sc. in Computer Engineering (Artificial Intelligence and Robotics)** — 2024 – Present
  Ferdowsi University of Mashhad — Mashhad, Iran
  - GPA: 3.69/4 (Graduated among **Top 3**)
- **B.Sc. in Computer Engineering** — 2020 – 2024
  Quchan University of Technology — Quchan, Iran
  - GPA: 3.73/4 (Graduated **1st Rank**)

Publications

- **Parsa Sinichi**, Miguel O. Bernabeu, Malihe Javidi. ***Comparison of Classical and Deep Learning-Based Feature Representations for Age-Related Macular Degeneration***. **MICAD 2024**
  10.1007/978-981-96-3863-5_46
- Fatemeh Naserizadeh, Erfan Akbarnezhad Sany, **Parsa Sinichi**, Seyyed Abed Hosseini. ***YOLOatt-Med: YOLO-Based Attention Mechanism for Medical Image Classification***. **ICCKE 2024**
  10.1109/ICCKE65377.2024.10874737
- Erfan Akbarnezhad Sany, Fatemeh Naserizadeh, **Parsa Sinichi**, Seyyed Abed Hosseini. ***Camouflage Object Segmentation with Attention-Guided Pix2Pix and Boundary Awareness***. **ICCKE 2024**
  10.1109/ICCKE65377.2024.10874655

Work Experience

- **C1Tech** — September 2024 – Present
  AI R&D Engineer (Full-time) — Mashhad, Iran
  A company with a leading position in clean room and hospital infrastructure in Iran
  - Contributed to releasing one of the few open-source Persian speech-to-text models, outperforming OpenAI's largest model on Persian ASR benchmarks. View Model on Hugging Face View Persian ASR Leaderboard
  - Fine-tuned large language models such as Qwen3 and Gemma3 for intent detection and slot filling, improving accuracy by 15% compared to baseline models
  - Helped develop a robust data collection pipeline for gathering and curating speech data
  - Worked with various ASR models (Whisper, Conformer) and TTS models (Tacotron2, FastSpeech2)
- **Teaching Assistant** — 2026
  Ferdowsi University of Mashhad
  - Assisted in teaching:
    - Machine learning (Graduate Course)
- **Teaching Assistant** — 2021 – 2024
  Quchan University of Technology — Quchan, Iran
  - Assisted in teaching:
    - Data Structures
    - Introduction to Programming
    - Signals and Systems

Research Projects & Relevant Experience

- **Multi-View Learning for Diabetic Retinopathy Grading under Missing Modalities** 2025-Present

  Supervisor: Dr. Yuting He (Google Scholar)

  - Designing a multi-view Diabetic Retinopathy grading framework that integrates multiple retinal views via a learnable fusion token and attention-based feature aggregation
  - Developing view-specific and shared representation learning using decoupled loss functions, explicitly separating individual view features from cross-view shared embeddings
  - Incorporating student–teacher knowledge distillation to enable token reconstruction and modality-robust inference, improving performance when one or more views are missing at test time

- **Comparison of Classical and Deep Learning-Based Feature Representations for Age-Related Macular Degeneration (Bachelor's Thesis)** 2024

  Supervisor: Prof. Malihe Javidi (Google Scholar)

  - Compared classical feature extraction methods (DWT, LBP, GLCM) with deep learning and foundation models for Age-related Macular Degeneration (AMD) classification using fundus images
  - Utilized foundation models (RETFound, RETFound-Green) trained on domain-specific retinal data and VGG16 pre-trained on ImageNet as feature extractors, highlighting the importance of relevant domain-specific training data
  - Achieved competitive performance on the ADAM dataset with a single fine-tuned RETFound model, outperforming complex ensemble and multi-model state-of-the-art techniques

- **Camouflage Object Segmentation with Attention-Guided Pix2Pix and Boundary Awareness** 2024
  - Developed an advanced method for camouflage object segmentation using a Pix2Pix architecture
  - Integrated a VGG16 backbone for feature extraction and Convolutional Block Attention Modules (CBAM) to enhance focus on critical image details
  - Implemented boundary loss to improve the model's ability to segment and differentiate objects, achieving superior performance on the CAMO dataset

- **YOLO-Based Attention Mechanism for Medical Image Classification (YOLOatt-Med)** 2024
  - Developed a medical image classification model using the YOLOv8 architecture for respiratory disease diagnosis
  - Integrated the Convolutional Block Attention Module (CBAM) to enhance feature extraction and improve discrimination between healthy and diseased tissues
  - Achieved superior performance on two distinct medical datasets, surpassing the majority of existing models

- **Raspberry Pi 4-Powered Control System for a Lower-Limb Robotic Exoskeleton** Summer 2023

  Supervisor: Prof. Alireza Akbarzadeh Tootoonchi (Google Scholar)

  - Developed control software for the exoskeleton using Raspberry Pi 4, integrating IMU and load-cell data via I2C for gait control
  - Implemented both the backend using WebSocket and the mobile application frontend, enabling real-time control, monitoring, and data logging
  - Designed feedback and monitoring modules to support clinical operation and improve usability for participants and researchers

- **Data Logging System for EMG, IMU, and FSR Sensors** 2023

  Supervisor: Prof. Alireza Akbarzadeh Tootoonchi (Google Scholar)

  - Built a Raspberry Pi-based logging system and application for EMG, IMU, and FSR sensors using CAN communication
  - Implemented synchronization and data-stream alignment methods to achieve stable 200 Hz recording
  - Implemented a real-time sensor status tracking and event reporting system, propagating connectivity information through the data pipeline to ensure stable and complete data collection

- **Computer-Vision Based System for Cloth Identification and Classification** 2022
  - Collected custom image data and labeled images using Labelme
  - Used OpenCV for image preprocessing and applied YOLO segmentation for segmenting clothes in real-world pictures

- Employed a Siamese Neural Network and Contrastive Learning to measure similarity between images for cloth identification

- **Real-Time 3D Hand Tracking Using Computer Vision and Unity** 2021 – 2022
  - Implemented a hand-tracking algorithm using MediaPipe in Python from a video feed
  - Established message passing between Python and Unity using WebSocket
  - Created a virtual 3D representation of the hand in Unity for interacting with objects in real time

SKILLS

- **Programming Languages & Frameworks:** Python, C++, TensorFlow, PyTorch, scikit-learn
- **Software:** Unity, ROS
- **Technical / Hardware:** Raspberry Pi, Surface EMG, IMU data collection, FSR sensors
- **Other:** Data synchronization, real-time systems, control algorithms, sensor integration

LANGUAGES

Persian: Native **English:** TOEFL : 109 (L:30,R:28,W:27,S:24)

AWARDS & HONORS

- **Winner, Barcode Programming Competition**
  - First-place winner in the Barcode programming competition organized by Quchan University of Technology

REFERENCES

- **Dr. Malihe Javidi** Research Fellow on Computer Vision in Medicine, Centre for Medical Informatics, Usher Institute, The University of Edinburgh, UK | Assisant Professor in Faculty of Computer and Electrical Engineering, Quchan Universtiy of Technology Email: mjavidi@ed.ac.uk
- **Dr. Zahra Eskandari** Assisant Professor in Faculty of Computer and Electrical Engineering, Quchan Universtiy of Technology Email: z.eskandari@qiet.ac.ir